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عبدالله السالم
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Project Course
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Economic Analysis and GDP Prediction

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1. Introduction

This project investigates global macroeconomic patterns from **2010 to 2023** using a curated dataset of international economic indicators. The study aims to understand how different countries' economies evolve over time and what factors may shape their growth trajectories. By examining a diverse group of countries including advanced, emerging, and resource-driven economies, the project highlights how structural differences, economic policies, and external shocks can influence national performance.

The analysis draws on key indicators such as **GDP, GDP Growth, GDP per Capita, Inflation, Unemployment, Government Finance, and Public Debt**. These features allow us to explore not only the size of each economy but also the stability, sustainability, and underlying health of their economic systems. Instead of focusing on a single country, the project takes a broader comparative perspective aimed at identifying patterns that hold across nations as well as unique outliers that stand apart.

The central questions guiding this study include:

- How have major macroeconomic indicators changed across countries over time, and what patterns of stability or volatility can be observed?
- What relationships exist between key economic indicators, and how strongly do they explain variations in GDP across countries?
- How accurately can GDP be predicted using multiple economic indicators (excluding GNI), and what does this reveal about each predictor's contribution?
- Which countries demonstrate the strongest economic conditions for investment based on combined measures of growth, stability, and overall performance?

To address these questions, the project proceeds in several steps. First, the **Exploratory Data Analysis** examines the dataset, identifies irregularities, and visualizes historical trends. Next, **Comparative Visualizations** provide a side-by-side look at how countries differ in growth, wealth, and stability. The **Correlation Analysis** investigates statistical relationships to uncover deeper insights about how different indicators interact. Lastly, the **Predictive Modeling** component uses historical data to forecast future GDP values and evaluate the reliability of these predictions.

Together, these analyses offer a broad and data driven perspective on the economic trajectories of multiple nations, highlighting both shared global patterns and country specific dynamics.

2. Exploratory Data Analysis (EDA):

2.1 Data Selection

The analysis dataset was constructed by selecting a group of European countries along with several major global economies. Additional non-European economies include Canada, the United States, China, Japan, Korea, and the Russian Federation. The included countries comprise a mix of advanced, emerging, large, and small economies. These indicators were selected because they jointly capture economic size, stability, labor market conditions, and fiscal capacity. Together, they provide a multidimensional view of macroeconomic performance rather than relying on GDP alone. The European group contains nations such as Austria, Belgium, France, Germany, Netherlands, Spain, Sweden, Switzerland, the United Kingdom, and others. Additional non-European economies include Canada, the United States, China, Japan, Korea, and the Russian Federation.

This diverse selection enables a broad comparison across different economic structures.

2.2 Missing Data imputation

A detailed inspection of missing values was conducted at both the country level and the feature level.

Missing values by country:

country_name	
Montenegro	56
San Marino	39
Japan	36
Germany	31
Serbia	30
Poland	28
Luxembourg	28
Finland	28
Denmark	28
Latvia	28
Cyprus	28
China	28
Slovenia	28
Portugal	28
Malta	24
Netherlands	24
North Macedonia	20

Missing values by feature:

Interest Rate (Real, %)	335
Public Debt (% of GDP)	274
Government Expense (% of GDP)	42
Government Revenue (% of GDP)	41
Tax Revenue (% of GDP)	41
Unemployment Rate (%)	16
Current Account Balance (% GDP)	8
Gross National Income (USD)	8
Inflation (CPI %)	2
GDP (Current USD)	1
GDP per Capita (Current USD)	1
Inflation (GDP Deflator, %)	1
GDP Growth (% Annual)	1
country_name	0
country_id	0
year	0
dtype: int64	

2.3 Summary Statistics of Main Features

Feature	Mean	Std	Min	Max
Inflation (CPI %)	3.26	5.07	-2.10	59.22
GDP (Current USD)	1.36e+12	3.52e+12	1.42e+9	2.77e+13
GDP per Capita (USD)	33,229	26,043	2,094	133,712
Unemployment Rate (%)	8.45	5.65	0.79	33.14
GDP Growth (% Annual)	2.16	3.87	-28.75	24.61
Government Expense (% of GDP)	35.62	8.56	16.07	66.46
Government Revenue (% of GDP)	33.37	7.86	10.96	56.53

The main observations from the statistics indicate that there is a very wide GDP range, reflecting strong heterogeneity among the selected economies. Inflation exhibits a right skewed distribution, with a few years experiencing extreme price instability. GDP growth displays high volatility, which is consistent with global business cycles and country-specific economic shocks. Meanwhile, government finance indicators show relatively stable central tendencies across countries.

2.4 Data Cleaning

After completing the exploratory analysis, several cleaning procedures were applied to ensure that all economic indicators were consistent, reliable, and suitable for modeling and visualization.

2.4.1 Handling Missing Values

A two-step imputation strategy was employed:

- Time-Series Completion:

For features with only 1–2 missing values within a country’s time series, missing entries were filled using forward fill and backward fill methods. This preserved the temporal continuity of each indicator.

- KNN Imputation:

For countries with more substantial missing data, each was assigned a set of economically similar neighbouring countries. Missing values were replaced with the yearly average of the

same indicator across these neighbours. This approach maintained realistic regional patterns and prevented the introduction of artificial bias.

2.4.2 Cleaning Decisions

To reduce noise and prevent unreliable modeling outcomes, variables with a high proportion of missing values were removed from the dataset. These variables included:

Interest Rate (Real, %), Public Debt (% of GDP), and country_id

Removing these features improved data quality and ensured that the remaining indicators could be used reliably in comparative and predictive analyses.

2.4.3 Log Transformation of Wealth Indicators

To address skewness and high variance in wealth related variables, the GDP, Gross National Income, and GDP per Capita were transformed using the natural logarithm. This transformation reduces the influence of extreme values, stabilizes variance, and facilitates interpretation in comparative and predictive analyses. New columns containing the log transformed values were added to the dataset for subsequent modeling and visualization.

2.4.4 Outlier Detection

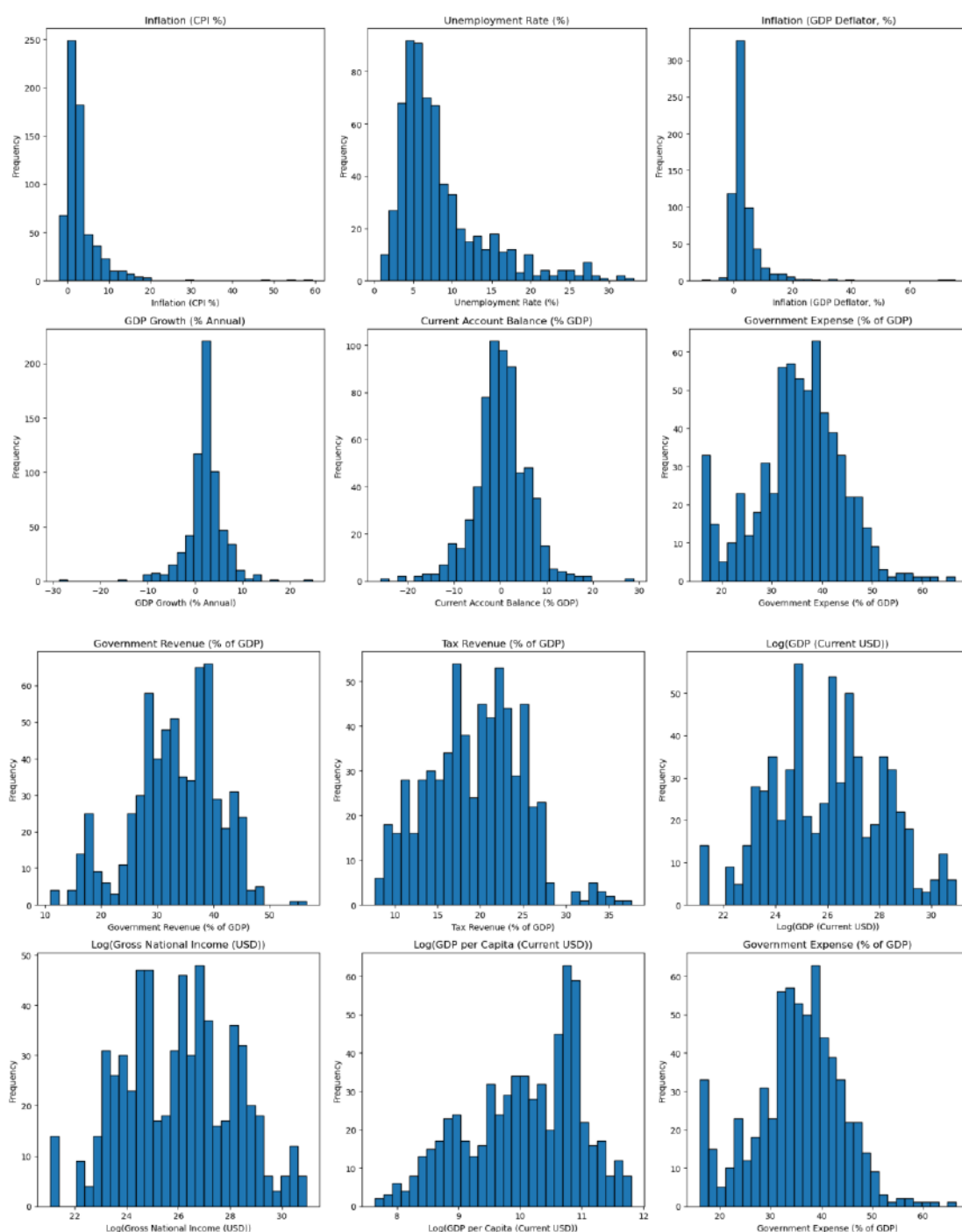
Potential outliers in all numeric variables (excluding year) were identified using the Interquartile Range (IQR) method. The number of outliers per variable was counted to assess unusually high or low values. These outliers reflect real economic phenomena rather than systematic errors, and therefore, they were retained in the dataset. Identifying them helps guide data transformations or cleaning decisions while preserving meaningful variation for comparative and predictive analyses.

```
'Inflation (CPI %)': 61
'GDP (Current USD)': 89
'GDP per Capita (Current USD)': 19
'Unemployment Rate (%)': 45
'Inflation (GDP Deflator, %)': 52
'GDP Growth (% Annual)': 57
'Current Account Balance (% GDP)': 27
'Government Expense (% of GDP)': 6
'Government Revenue (% of GDP)': 5
'Tax Revenue (% of GDP)': 4
'Gross National Income (USD)': 91
'Log(GDP (Current USD))': 0
'Log(Gross National Income (USD))': 0
'Log(GDP per Capita (Current USD))': 0
```

2.4.5 Normality Analysis of Key Economic Variables

The distributions of the six economic indicators show mixed patterns of normality. Inflation and Inflation based on the GDP Deflator both display strong right skewness, with most countries experiencing low inflation and a small number showing extreme spikes, making these variables far from normally distributed. The Unemployment Rate is also not normal, as it is

right-skewed and lacks a symmetric bell-shaped pattern. GDP Growth shows the closest resemblance to a normal distribution, with most values centered between 0–5%, although the presence of both negative contractions and high positive growth creates some deviation from perfect normality. The Current Account Balance is relatively more balanced around zero, making it closer to normal than the other indicators, though slight skewness remains. Government Expense, however, is not normally distributed due to its right skewed shape and long tail of high spending countries. Overall, only GDP Growth and the Current Account Balance show near-normal patterns, while the other indicators significantly deviate from normality.



3. Comparative Visualization

Average GDP Growth per Country

To evaluate long-term economic expansion, the average GDP Growth was calculated for each country over the observed period.

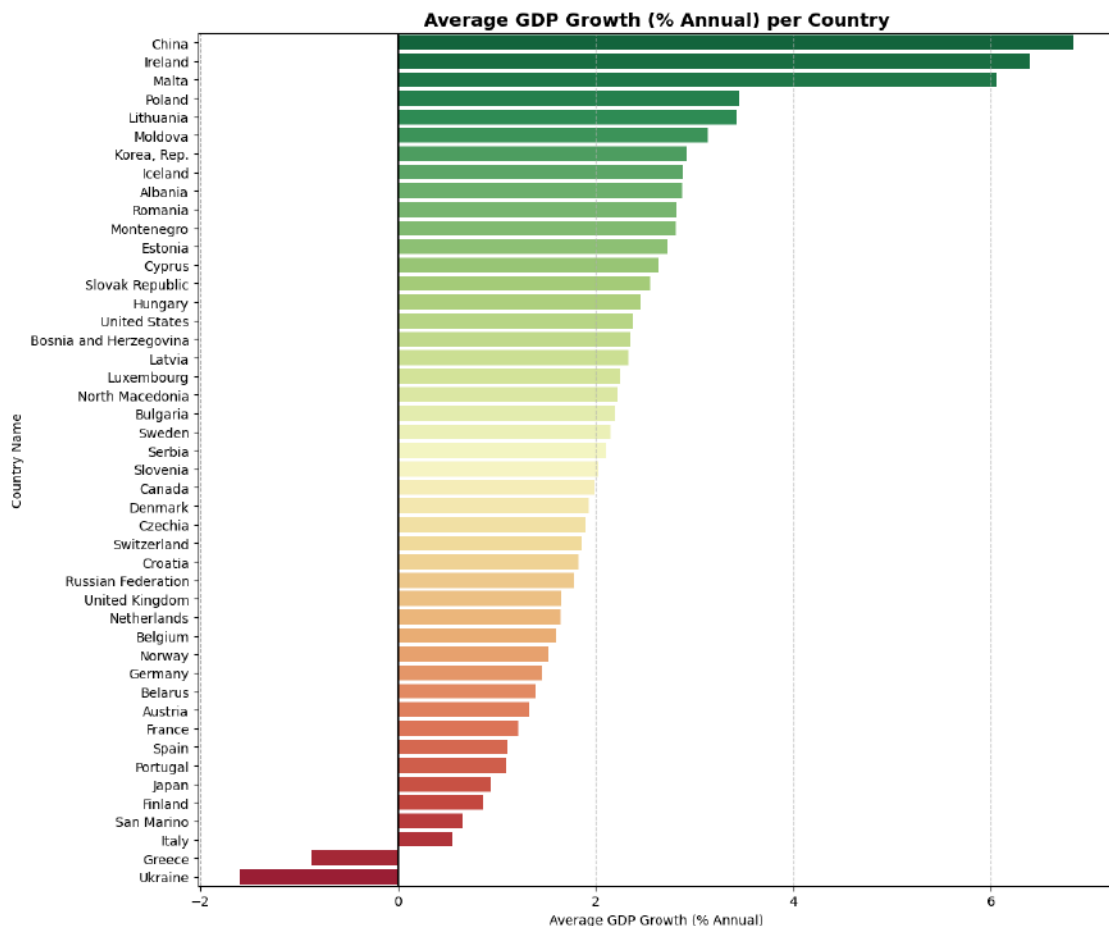
Key Findings:

High-Growth Countries: China, Ireland and Malta lead with growth rates above 6%.

Moderate Growth Countries: Countries such as the United States, Slovak Republic, and Hungary show steady growth (2–2.5%).

Low-Growth: Italy, San Marino, and Finland are near stagnation (~ 1%), while Greece shows moderate contraction and Ukraine the largest decline (~-2%).

This analysis highlights disparities in economic performance, showing which countries experience rapid expansion and which face stagnation or contraction, providing insight into regional economic resilience and development trajectories.



Country Economic Performance: Risk vs. Reward (GDP Growth)

This analysis highlights the trade-off between growth and stability, showing which economies are most favourable for investment based on growth potential and risk tolerance.

Country-level economic performance was evaluated using average GDP Growth of all countries in the sample as Reward. If a country's average GDP Growth is higher than the overall mean of GDP Growth, it indicates that the country achieved relatively strong economic expansion over the observed period. Conversely, if its growth is lower than the mean, it indicates weaker performance compared to the global sample.

Risk was measured using the standard deviation of GDP Growth, which captures volatility. A country with a higher standard deviation than the sample average standard deviation is considered less stable, experiencing large fluctuations in growth over time. A lower standard deviation than the average indicates more stable growth in previous years.

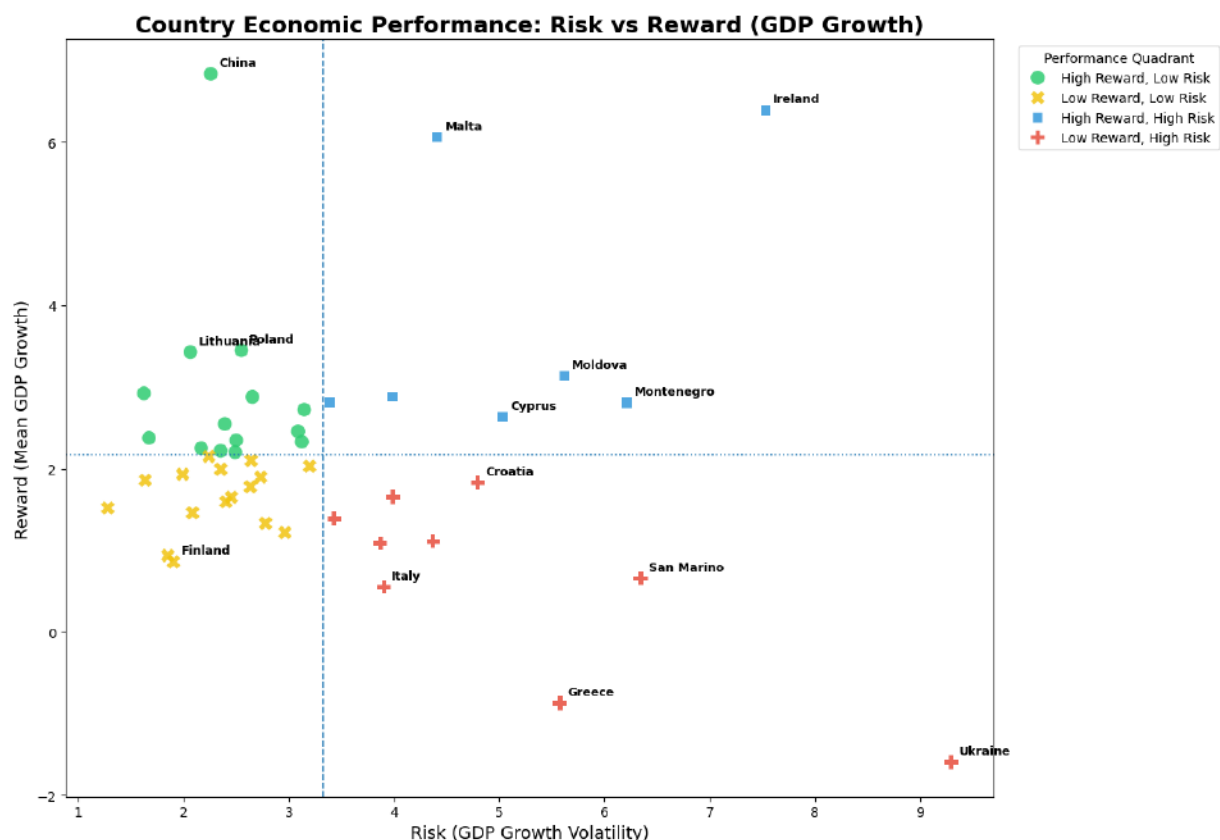
Key Findings:

High Reward, Low Risk (Ideal): China, Lithuania – strong growth with low volatility.

High Reward, High Risk (Speculative): Ireland, Malta – high growth but unstable.

Low Reward, Low Risk (Stable but Slow): Finland – low but steady growth.

Low Reward, High Risk (Challenging): Ukraine, Greece – low growth and high volatility.

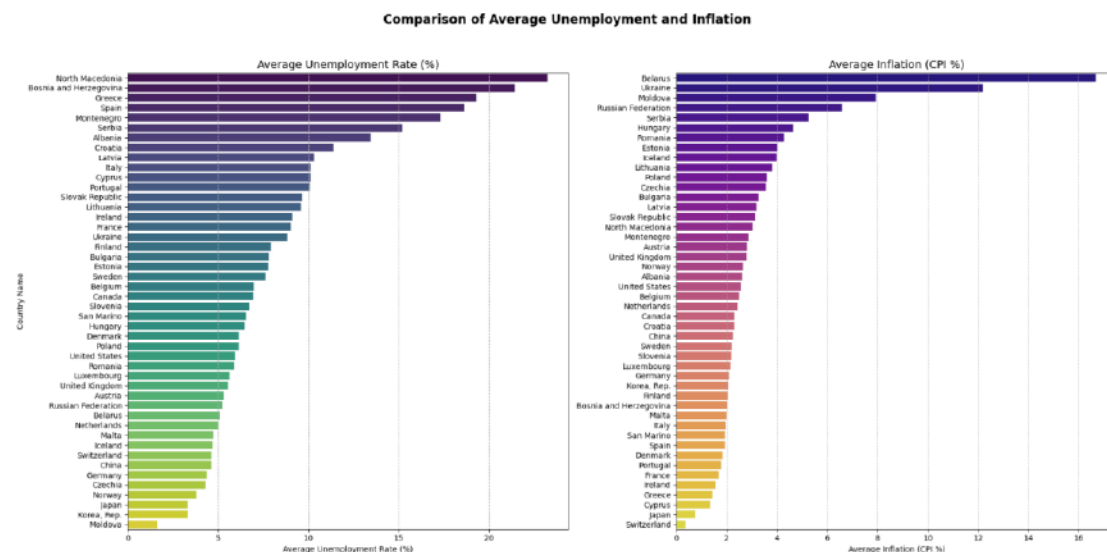


Average Unemployment and Inflation

To assess long-term labor market and price stability conditions, the average Unemployment Rate (%) and average Inflation (CPI %) were calculated for each country over the observed period.

Key Findings:

- Countries with higher average unemployment such as North Macedonia face persistent labor market pressures, while those with lower unemployment show relatively stable employment conditions.
- Countries with higher average inflation like Belarus and Ukraine experience greater price instability, whereas countries with lower inflation like Switzerland and Japan reflect more stable purchasing power.



This analysis complements the Misery Index by separating its two components, providing a more granular view of economic discomfort and helping to identify the specific source unemployment or inflation of cross country differences.

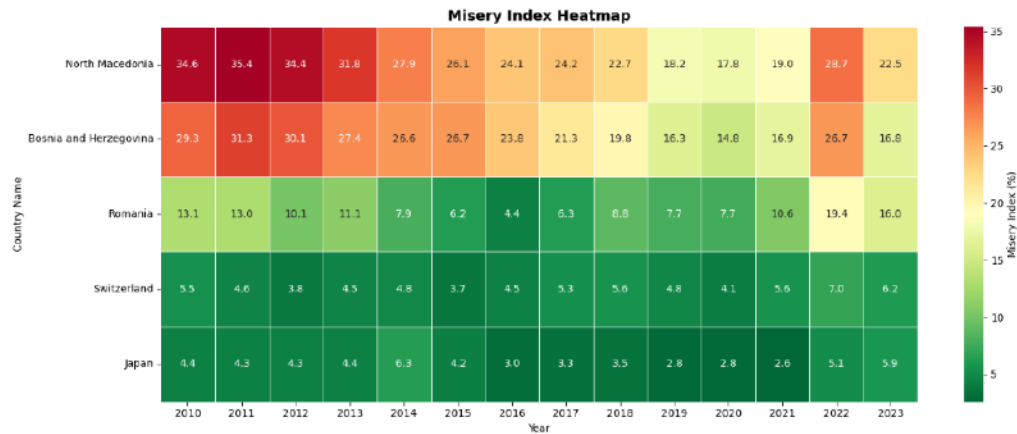
Misery Index

The Misery Index was calculated for each observation by summing the Unemployment Rate and Inflation to assess economic discomfort across countries. The average index was then computed for each country, and five representative countries were selected for visualization: the two with the highest index, the two with the lowest index, and one near the overall median. A heatmap was generated to display year by year Misery Index values, highlighting temporal trends and cross country differences.

Key Findings:

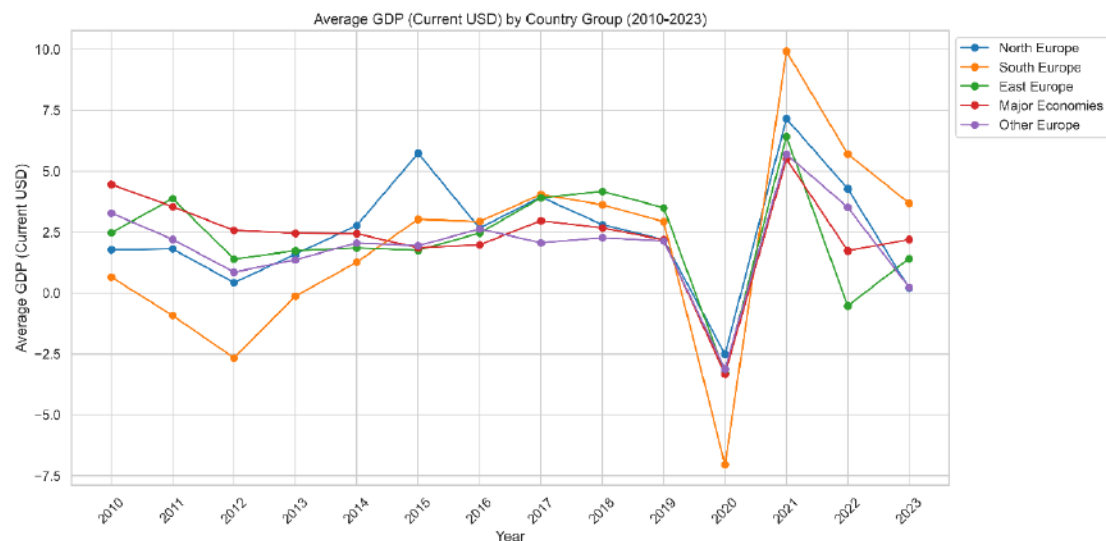
- North Macedonia had the highest average Misery Index, indicating the greatest combined economic pressure from unemployment and inflation.
- Japan had the lowest average Misery Index, reflecting consistently low and stable unemployment and inflation.

The heatmap demonstrates fluctuations in economic hardship over time, providing a clear comparative perspective on well-being across countries.



Trends in Average GDP Across Country Groups

The plot of average GDP by country group reveals several key trends. All country groups experienced a similar general trend: a period of relative stability or modest growth between 2010 and 2019, followed by a dramatic and sharp drop in 2020. This massive decline, likely attributable to the global economic shock of the COVID-19 pandemic, was universally severe, with South Europe recording the lowest point (around -7.5). Following the 2020 trough, all groups saw an extremely sharp rebound in 2021. North Europe and Major Economies generally maintained the highest average GDP levels throughout the period, with North Europe peaking highest in both 2015 and the 2021 rebound (around 10.0). The rebound in 2021 was the most significant single-year change observed. In the final years, from 2021 to 2023, most groups show a moderate decline or stabilization in their average GDP.



Post-COVID GDP Recovery

This visualization provides a clear comparison of recovery trajectories, complementing other indicators like Government Finance and the Misery Index, and highlighting how different countries respond to global economic shocks.

To assess post-pandemic economic performance, a GDP Index was calculated for each country, setting 2019 as the base year (2019 = 100). The base year serves as a reference point, meaning that the GDP in 2019 is normalized to 100 for all countries. This allows the index expresses

each year's GDP as a percentage of the 2019 level, allowing comparison of recovery speed across countries regardless of absolute GDP size.

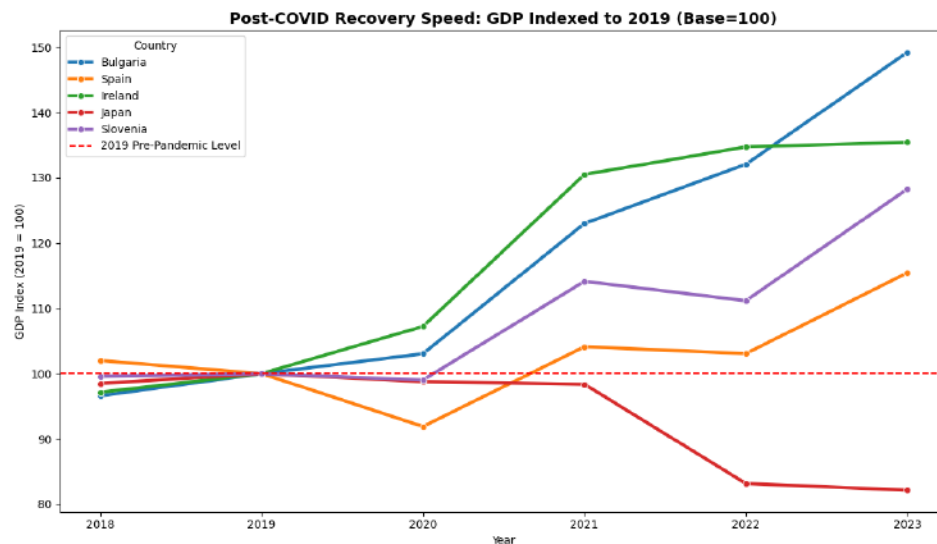
$$GDP\ Index_{c,t} = \frac{GDP_{c,t}}{GDP_{c,2019}} \times 100$$

c: country, t: year, GDP index for country c in year t.

The analysis focused on 2020 onward, highlighting five representative countries: the two with the fastest recovery, one near the median recovery, and the two with the slowest recovery. A line chart was used to visualize the GDP Index over time, with a reference line at 100 representing the pre-pandemic level.

Key Findings:

- Bulgaria achieved the fastest recovery, surpassing pre-pandemic GDP levels within a few years, indicating a strong economic rebound.
- Japan experienced the slowest recovery, with GDP remaining below 2019 levels. This slow growth is linked to structural factors present even before the pandemic, such as an aging population, weak domestic consumption, and long-term economic stagnation.



Comparative Investment Analysis

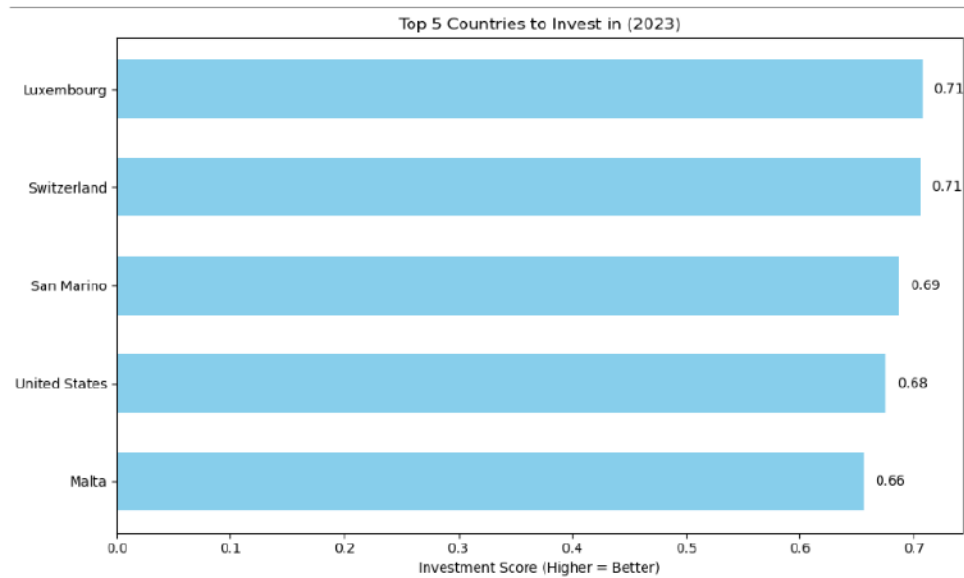
An Investment Score was computed for each country using key economic indicators: GDP per Capita, GDP Growth, and Inflation (CPI %). GDP per Capita and GDP Growth were normalized to a 0–1 scale, while Inflation was inversely normalized to reflect the preference for lower inflation. The three normalized values were averaged to produce a single Investment Score, where higher scores indicate more favourable conditions for investment.

Countries were ranked by their Investment Score for the most recent year, and the top 5 countries were visualized in a horizontal bar chart.

Key Findings:

- Luxembourg ranked highest, offering the most attractive combination of high wealth, strong economic growth, and low inflation.
- Switzerland, San Marino, the United States, and Malta followed, showing slightly less favourable conditions for investment.

This approach provides a clear, quantitative comparison of countries, balancing multiple economic indicators to guide potential investment decisions.



Government Revenue Composition Analysis

The composition of government revenue was analyzed for each country, separating tax revenue from other sources of revenue, to understand how governments fund their operations and to highlight differences in fiscal structure.

For this analysis, the average Tax Revenue (% of GDP) and Government Revenue (% of GDP) were calculated for each country. Other Revenue was obtained by subtracting tax revenue from total government revenue. A stacked bar chart was used to show each country’s revenue composition: the lower portion of the bar represents tax revenue, and the upper portion represents other revenue. Colors were used to make the comparison across countries clear.

Key Findings:

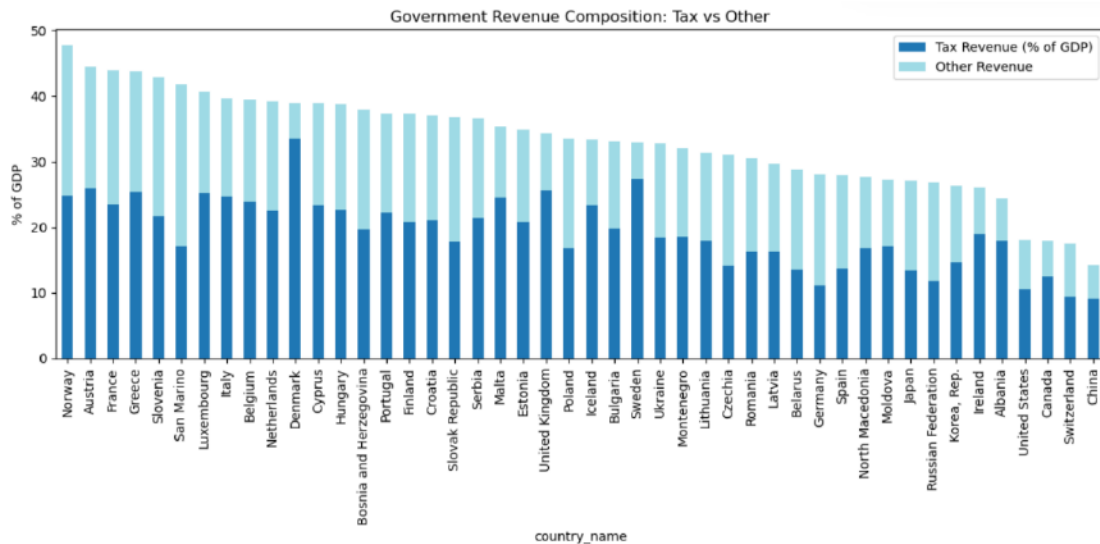
Denmark has the highest proportion of tax revenue, indicating strong reliance on taxes.

China and Switzerland have lower tax revenue proportions, relying more on other sources of income.

Countries with higher tax revenue proportions tend to have more predictable funding, while those relying on non-tax revenue may be more exposed to fluctuations in alternative income sources.

Implications:

The analysis highlights differences in fiscal strategies across countries and provides insight into government financial sustainability.



4. Correlation Analysis

A correlation matrix was computed for all numeric macroeconomic indicators to quantify the strength and direction of linear relationships between variables. To improve readability, long column names were shortened, and the results were visualized using a heatmap with annotated correlation coefficients.

Key Findings:

- Gross National Income (GNI) and GDP: A very strong positive correlation was observed between Gross National Income and GDP.
- Government Indicators: Good correlations were found between Government Expense and both Government Revenue and Tax Revenue
- Other Features: Most other features exhibited no or only moderate correlations.

Interpretation:

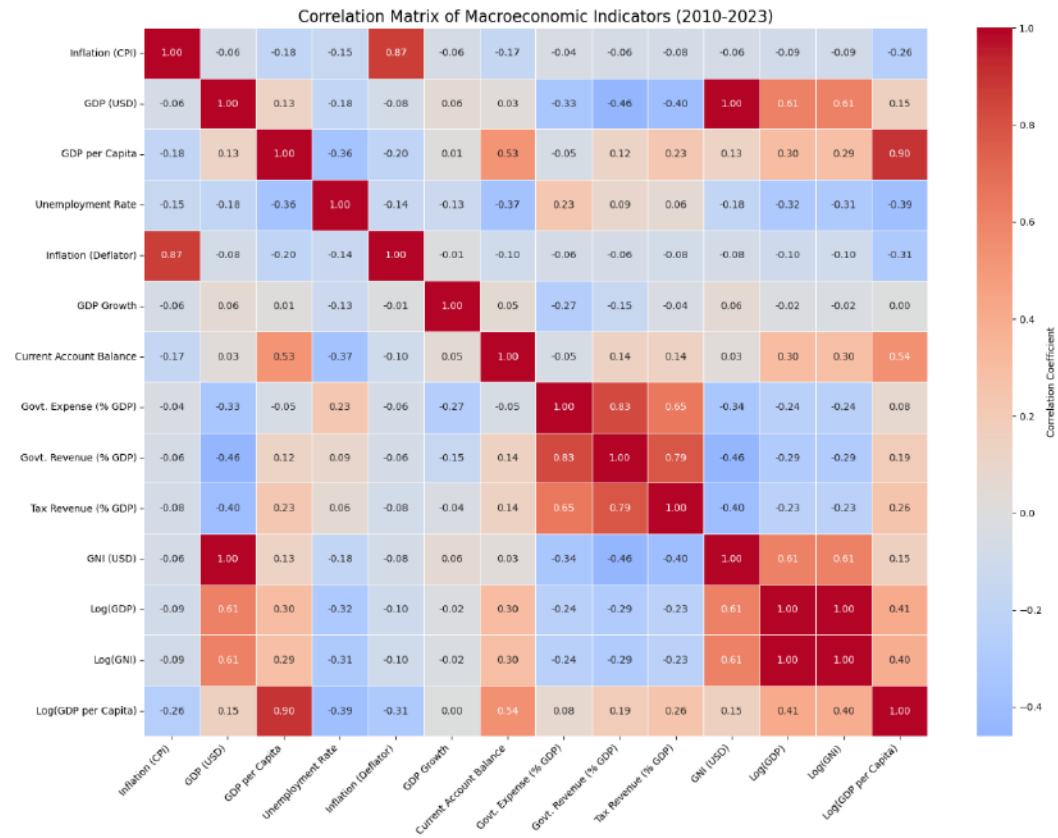
GDP measures the total market value of all goods and services produced within a country, while GNI adjusts GDP by adding income earned by residents from abroad and subtracting income earned by non-residents within the country. Because most countries' net income from abroad is relatively small compared to domestic production, GNI closely follows GDP. Predicting GDP solely from GNI is not meaningful because the two are mechanically linked, GNI is calculated from GDP with minor adjustments. Any model based on this relationship largely reproduces an accounting identity plus noise, providing little new economic insights.

Good correlations between government expenditure, revenue, and tax revenue reflect consistent fiscal patterns, as higher government spending is typically matched by higher revenue and taxation. Other features' weak or moderate correlations suggest they are more independent in their behavior.

Implications:

- The high GDP–GNI correlation suggests redundancy in including both indicators in certain analyses or predictive models unless international income is specifically of interest.
- Strong correlations among government indicators confirm the dataset's internal consistency and expected economic relationships.

- Weak correlations among other indicators highlight the need to consider each variable separately in modeling and analysis.



5. Predictive Modeling

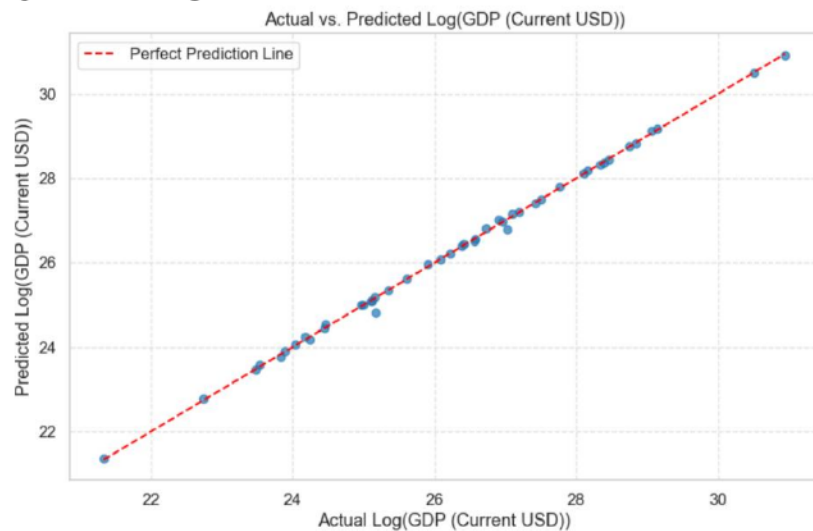
Linear Regression was selected as a simple and highly interpretable baseline model to provide a clear quantitative assessment of the relationship between macroeconomic features and GDP before considering more complex, non-linear approaches. The model was used to predict GDP for 2023 under two experimental scenarios.

In both scenarios, a broad set of fundamental macroeconomic indicators was considered, including measures of economic momentum (GDP Growth), price stability (Inflation), labor market conditions (Unemployment Rate), external balance (Current Account Balance), and fiscal policy (Government Expense). The two scenarios differ only in whether Gross National Income (GNI) is included as a predictor.

Scenario 1 (with GNI):

This scenario yields extremely high performance, with an R-squared score: 0.9986, (MAE): 0.0426, (MSE): 0.0058, (RMSE): 0.0760. While these results appear excellent, they are economically misleading. As discussed in Section 4, GDP and GNI are mechanically linked through national accounting identities, with GNI being derived directly from GDP with minor adjustments. Consequently, the model does not learn meaningful economic relationships but

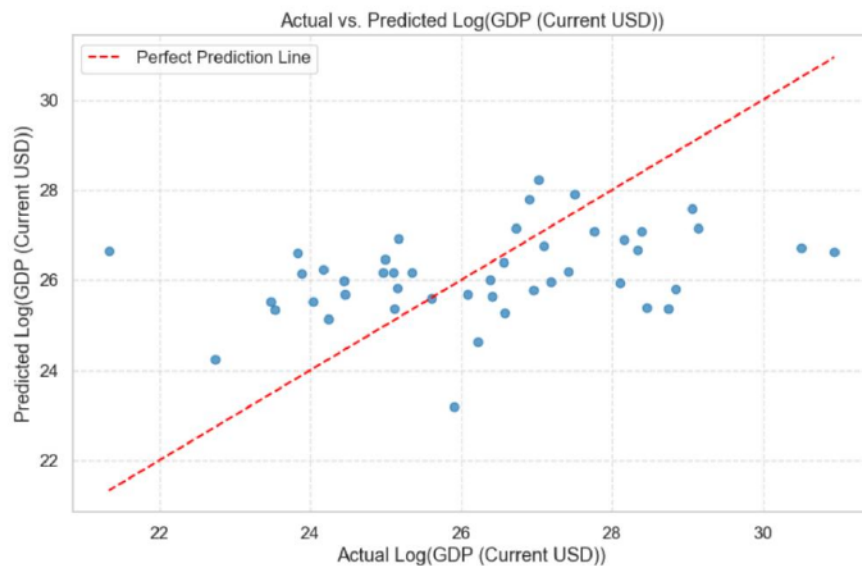
instead reproduces an accounting identity, resulting in artificially inflated performance rather than genuine predictive insight.



Scenario 2 (without GNI):

The model demonstrates weak accuracy, capturing general trends in GDP while highlighting the complexity of the underlying economic relationships. These factors primarily explain short-term deviations or rates of change, not the overall size of the economy and excluding GNI ensures the model focuses on the contribution of diverse macroeconomic indicators rather than reproducing an accounting identity.

R-squared score of 0.1075, meaning roughly 10.75% of the variance in GDP is explained by the included features, reflecting that the model does not capture the determinants of economic size. The Mean Absolute Error (MAE = 1.5758) and Root Mean Squared Error (RMSE = 1.9284) indicate weak deviations from actual values on the log scale.



Conclusion

After conducting extensive experiments with multiple model specifications and feature combinations, including the use of lagged macroeconomic indicators from the previous year ($t-1$) to predict GDP Growth in the current year (t), the results indicate that no model achieved strong or reliable predictive performance. While some configurations were able to capture broad directional trends, overall accuracy remained weak, reflecting the inherent complexity and volatility of macroeconomic dynamics. These findings suggest that short-term GDP growth is influenced by factors beyond the scope of the available indicators, including structural changes, external shocks, and policy interventions that are difficult to capture using historical macroeconomic data alone. Consequently, this study highlights the limitations of linear models and lag-based features in forecasting GDP growth and reinforces the importance of cautious interpretation when applying predictive modeling to macroeconomic outcomes.